

5. Instrumentation

└ 5.2 Alerting and metrics

└ 5.3.4 Kusto Query Language (KQL) basics

-
1. What is Kusto Query Language (KQL) and where is it used in Azure DevOps monitoring?
 2. What are the main KQL data tables relevant for Application Insights?
 3. How do you write a basic query to retrieve request telemetry in KQL?
 4. What is the syntax for filtering, projecting, and ordering data in KQL?
 5. How do you aggregate and summarize data using KQL?
 6. How can you join multiple tables in KQL for advanced insights?
 7. What functions are available in KQL to parse and extract fields from logs?
 8. How do you visualize query results in Azure Monitor using KQL?
 9. What are common pitfalls and best practices for writing efficient KQL queries?
 10. How do you save and share KQL queries in Azure Monitor or Application Insights?
-

1. What is Kusto Query Language (KQL) and where is it used in Azure DevOps monitoring?

KQL is a read-only query language for analyzing large volumes of structured, semi-structured, and unstructured data stored in Azure Data Explorer, Log Analytics, and Application Insights.

2. What are the main KQL data tables relevant for Application Insights?

Common tables: requests, dependencies, exceptions, traces, customEvents, and availabilityResults.

3. How do you write a basic query to retrieve request telemetry in KQL?

```
requests  
| take 10
```

4. What is the syntax for filtering, projecting, and ordering data in KQL?

Use where to filter, project to select columns, and order by to sort. Example:

```
requests  
| where resultCode == "500"  
| project timestamp, name, url  
| order by timestamp desc
```

5. How do you aggregate and summarize data using KQL?

Use summarize for aggregation. Example:

```
requests  
| summarize count() by resultCode
```

6. How can you join multiple tables in KQL for advanced insights?

Use the join operator. Example:

```
requests  
| join kind=inner (dependencies) on operation_Id
```

7. What functions are available in KQL to parse and extract fields from logs?

Functions: parse, extract, split, substring, tostring. Example:

```
traces  
| extend errorType = extract("Error: (.*)", 1, message)
```

8. How do you visualize query results in Azure Monitor using KQL?

Run the KQL query and select visualization options (time chart, pie, table) from the UI toolbar.

9. What are common pitfalls and best practices for writing efficient KQL queries?

- Pitfalls: Large scans, missing filters, unnecessary joins.
 - Best practices:
 - Filter early,
 - project only needed fields,
 - use summarization,
 - and test queries for performance.
-

10. How do you save and share KQL queries in Azure Monitor or Application Insights?

Use "Save" or "Pin to dashboard" in the query editor. Share query links or save as a shared workbook for collaboration.